Course Information

Andy Wang

COP 5611

Advanced Operating Systems

Contact Information

- Andy Wang (awang@cs.fsu.edu)
- Office: 264 Love Building
- Office hours: M 4 - 5pm
  F 4 – 5pm
  (also by appointments)
- Class website:
  http://www.cs.fsu.edu/~awang/courses/cop5611_s2010

Objectives

- Become exposed to classic and current operating systems literature
- Gain experience in doing OS research
- Develop projects that lead to publishable results

Prerequisites

- COP 4610 (operating systems)
- CDA 3101 (computer organizations)
- Knowledge of the UNIX environment
- Proficiency in C

Course Materials

- Lecture notes and papers (posted on the class website)
- No required textbooks

Recommended Textbooks

- Tanenbaum and Van Steen, *Distributed Systems Principles and Paradigms*
- Singhal and Shivaratri, *Advanced Concepts in Operating Systems*

Background Textbooks

- Tanenbaum, *Modern Operating Systems*
- Silberschatz, Galvin, Gagne, *Operating System Concepts*
- Nutt, Operating Systems: *A Modern Perspective*

Kernel-Hacking Aids

- Nutt, *Kernel Projects for Linux*
- Kernighan, Ritchie, *The C Programming Language*
- Maxwell, *Linux Core Kernel Commentary*
- Corbet, Rubini, and Kroah-Hartman, *Linux Device Drivers*

Grading

Paper summaries and critiques	5%
Project	40%
Peer evaluation of projects	5%
Exam 1	10%
Exam 2	10%
Final	30%

Critiques

- Ten one-page single-spaced critiques on recent papers (< 1 yr), from the following venues, or from other venues with prior approval:
  - Conferences: SOSP, OSDI, EuroSys, RTSS, HotOS, HotStorage, FAST, Usenix Annual Technical Conference, Sigmetrics, Usenix Security Conference, StorageSS

Critiques

- One due each week, both in class and through turnitin.com (via blackboard), for the first 10 weeks

Critiques

- Need to address the following:
  - Summary
    - Problems/existing & new approaches/results
  - Intriguing aspects of the paper
    - Observations/trends/assumptions/techniques
  - How can the research be improved?
    - Techniques/experiments/handling of corner cases and assumptions

Project

- You need to develop a project in teams of two or three
- Goal:
  - Publishable results

Types of Papers

- Survey papers
- Position papers
- Simulation papers
- Measurement papers
- System papers

Some Example Projects

- Feasibility of using sound cues for debugging operating systems
- Feasibility study of applying economic models for distributed resource management
- Feasibility study of life-long storage of sensory inputs

Weekly Project Reports

- Demonstrate steady progress
  - Papers read
  - Obstacles encountered
  - New ideas
  - Software pieces built
  - Experiments

Project Proposal

- Due on the 5th week
- 10-minute presentation
  - All team members are required to participate
- 2-page written proposal
  - Motivation
  - The state-of-the-art
  - Methodology
  - Expected results
  - Timeline

Project Proposal

Include:

- Some references
- Division of labor amongst teams

Project Presentation

- During the last two weeks of the course
- 15-20 minutes
- 15-page (max) written paper due by the last lecture (double column, single-space, 10-pt font)
- Critiques on two other projects, not including yours

Exams

- In-class and closed-book, unless specified otherwise
- Essays and short answers
- Open research questions